Signals and Systems

Computer Assignment 2

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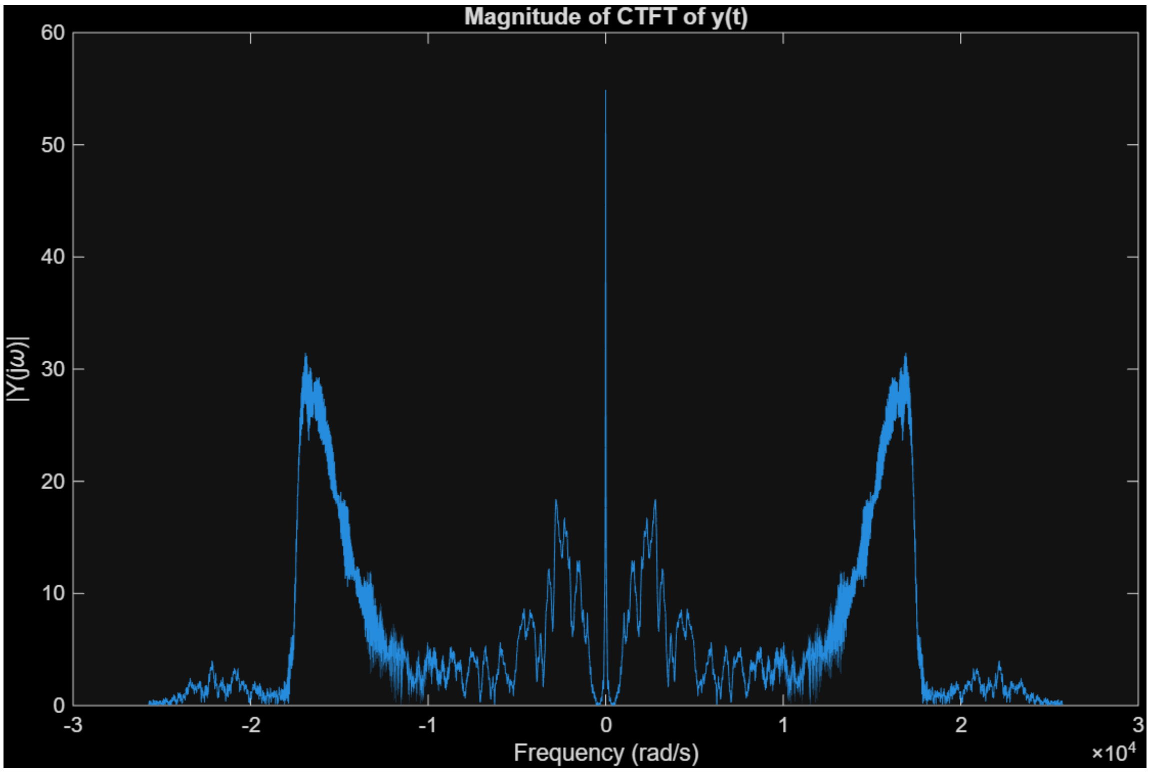
1. Properties of the Continuous-Time Fourier Transform

(a)



(b)

The audio heard in this section is the exact reverse of the original audio signal. From a theoretical standpoint, if $y(t) \leftrightarrow Y(j\omega)$, then for a real-valued signal, $y(-t) \leftrightarrow Y^*(j\omega)$. Therefore, taking the complex conjugate in the frequency domain corresponds directly to time reversal in the time domain.

(c)

We know that if a time-domain signal $y(t)$ is purely real, its Fourier transform must exhibit conjugate symmetry:

$$Y(-j\omega) = Y^*(j\omega) \tag{1}$$

This implies that the magnitude is an even function ($|Y(-j\omega)| = |Y(j\omega)|$) and the phase is an odd function ($\phi(-\omega) = -\phi(\omega)$).

For the signal $y_2(t)$:
Given $Y_2(j\omega) = |Y(j\omega)|$, we can evaluate it at $-j\omega$:

$$Y_2(-j\omega) = |Y(-j\omega)| = |Y(j\omega)| = Y_2^*(j\omega) \tag{2}$$

Since the condition of conjugate symmetry is met, its inverse Fourier transform, $y_2(t)$, is purely real.

For the signal $y_3(t)$:
Given $Y_3(j\omega) = e^{j\phi(\omega)}$, we evaluate it at $-j\omega$:

$$Y_3(-j\omega) = e^{j\phi(-\omega)} = e^{-j\phi(\omega)} = \left(e^{j\phi(\omega)}\right)^* = Y_3^*(j\omega) \tag{3}$$

This also satisfies the conjugate symmetry condition, meaning $y_3(t)$ is purely real.

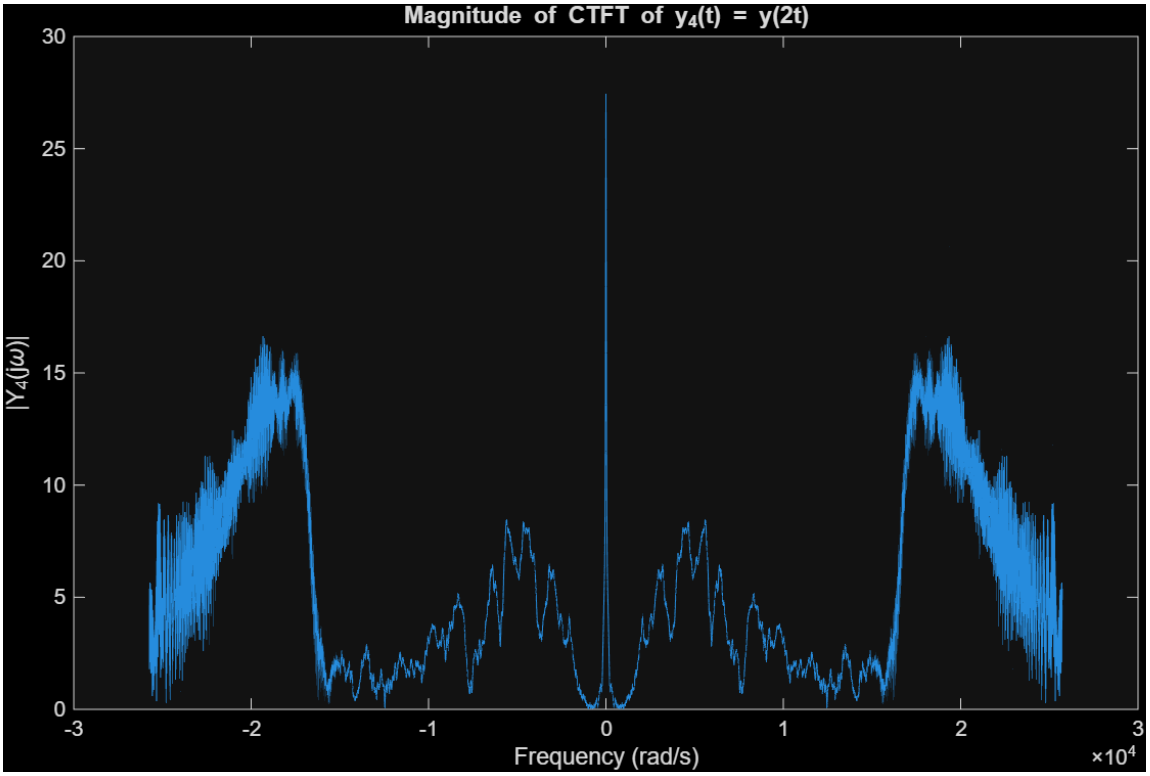
(d)&(e)

```
%part1_d&e
Y2 = abs(Y);
y2 = real(ifft(fftshift(Y2)));
sound(y2, fs);

pause(3);

Y3 = exp(1j * angle(Y));
y3 = real(ifft(fftshift(Y3)));
sound(y3, fs);
```

(f)

By listening to both signals, it becomes evident that the "phase-only" signal (y_3), although noisy, retains the overall structural characteristics and rhythm of the original sound, making it somewhat recognizable. In contrast, the "magnitude-only" signal (y_2) sounds like unintelligible, distorted noise. This demonstrates that in audio signals, the phase component carries significantly more vital structural information than the magnitude component.

(g)&(h)

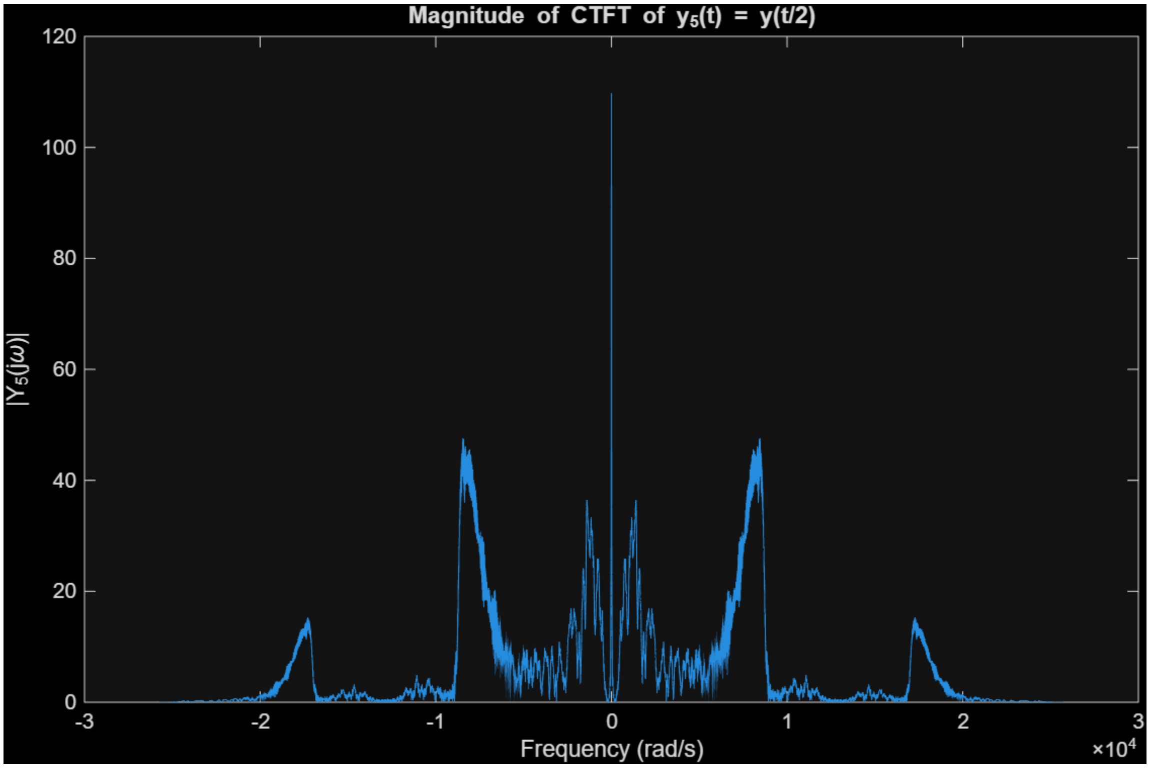


The generated audio plays twice as fast and sounds significantly higher in pitch. According to the time-scaling property of the Fourier Transform, time compression leads to frequency expansion:

$$y(at) \longleftrightarrow \frac{1}{|a|} Y\left(j\frac{\omega}{a}\right) \tag{4}$$

Since $a = 2$, the frequency spectrum expands, shifting the signal's energy to higher frequencies. This expansion manifests physically as a higher pitch.

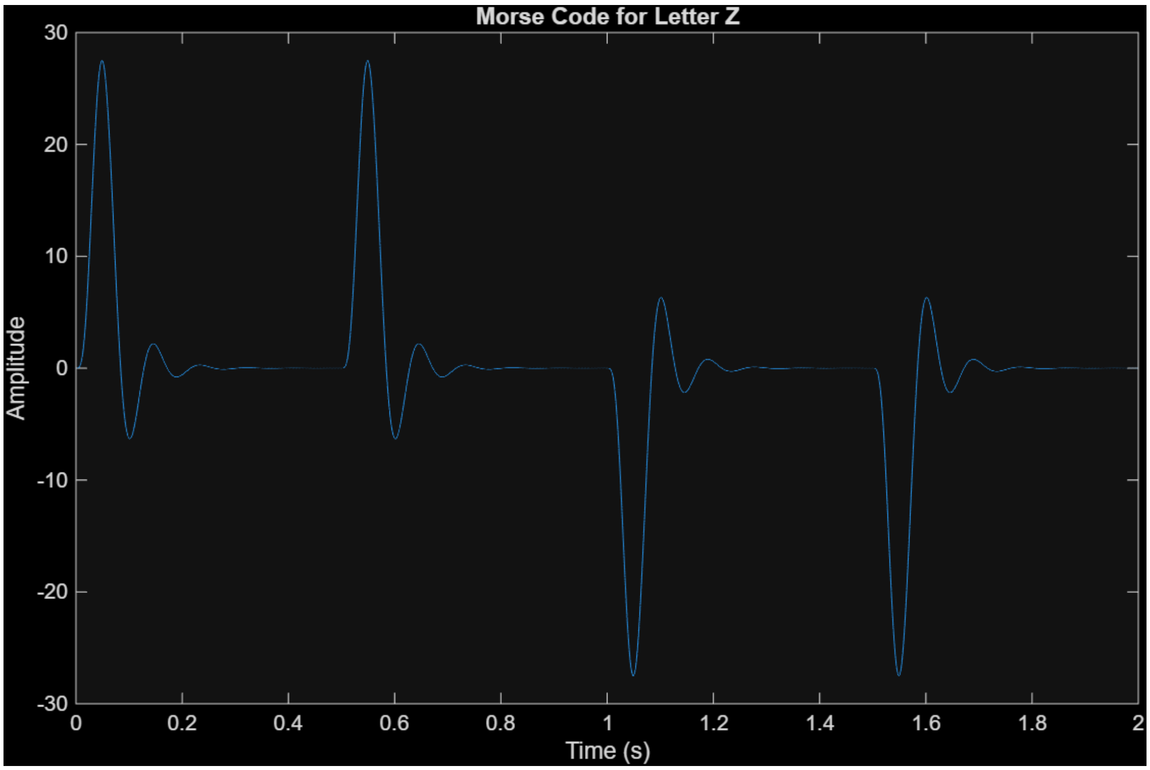
(i)&(j)&(k)



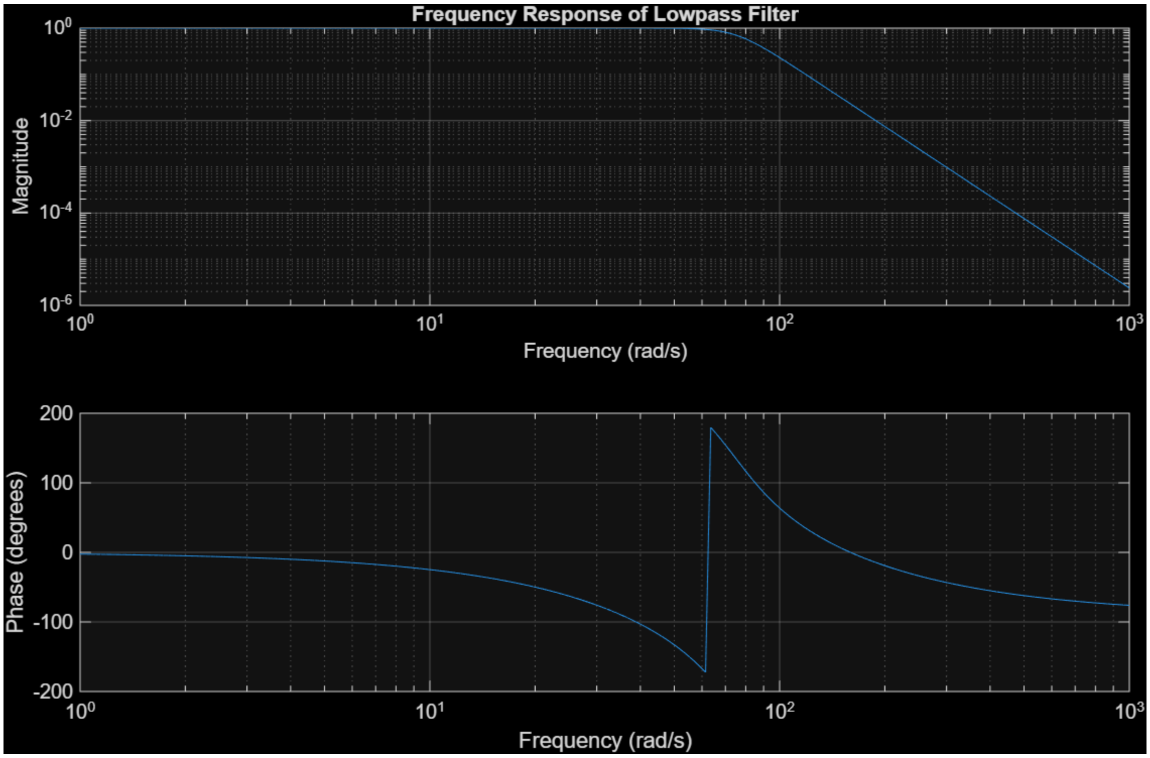
The audio plays slower and sounds much lower in pitch. Time expansion ($a = 1/2$) results in frequency compression. The bandwidth of the signal is effectively halved, which eliminates the higher-frequency components. The loss of these high frequencies results in a deeper, lower-pitched sound.

2. Amplitude Modulation and the Continuous-Time Fourier Transform

(a)

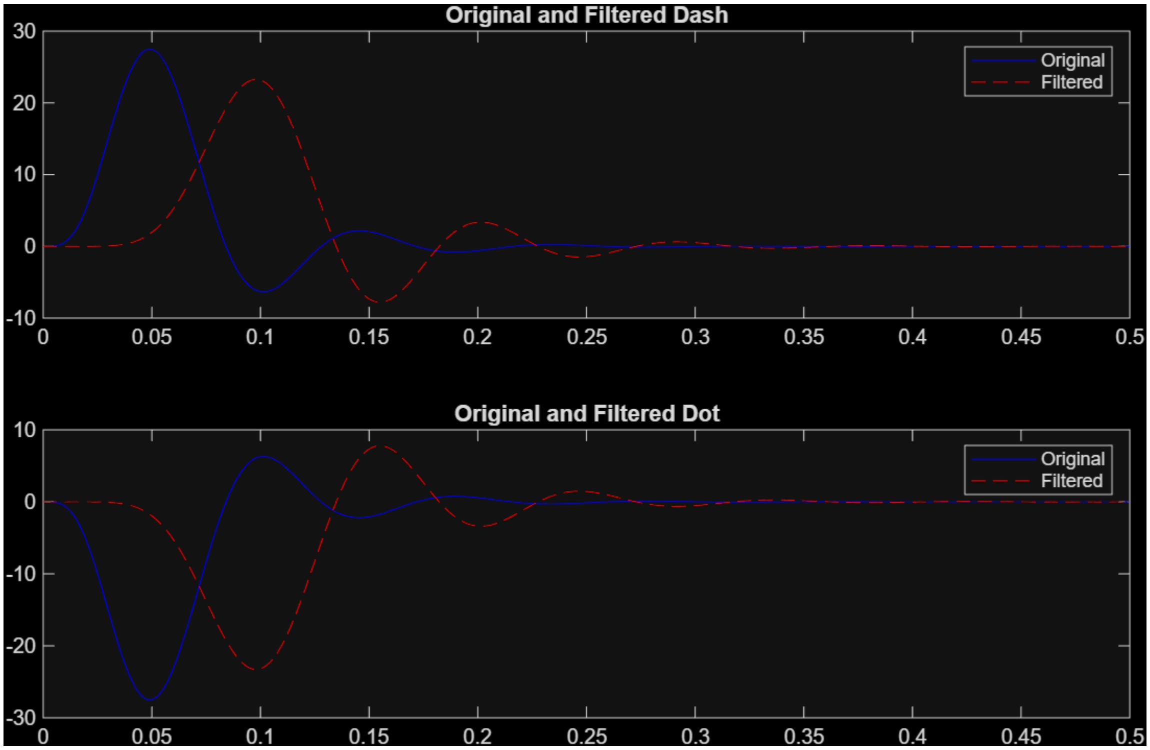


(b)



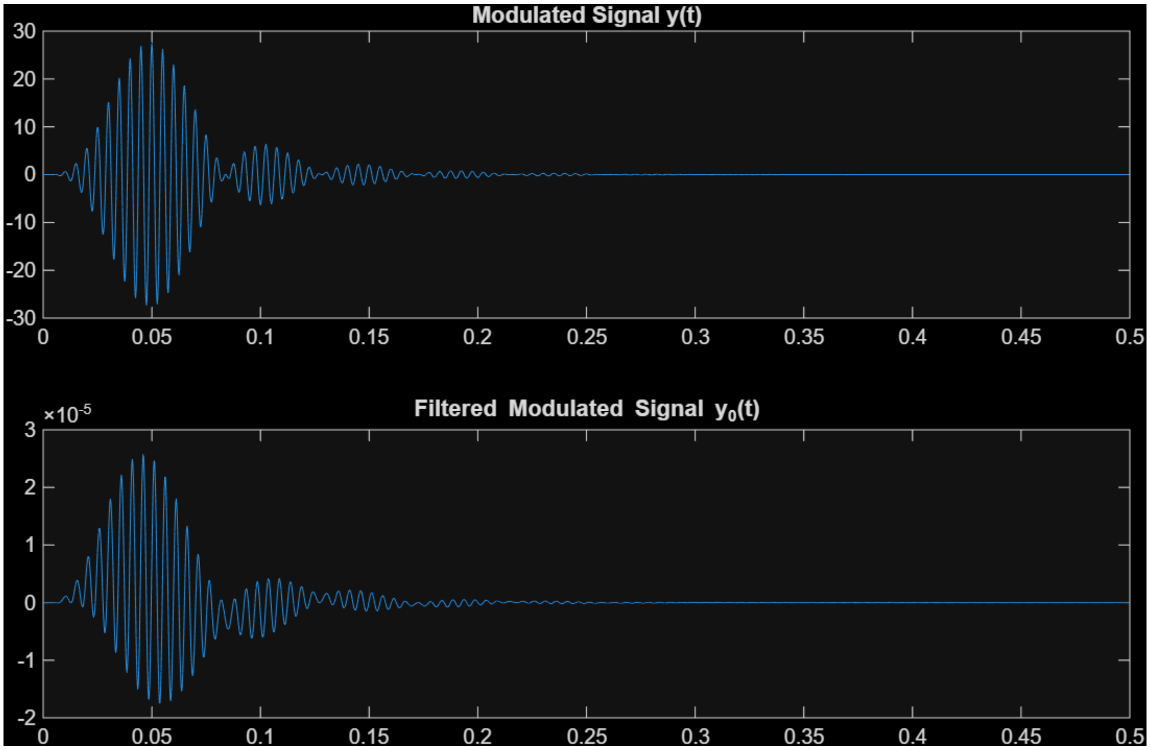
Frequency Response of Lowpass Filter
Magnitude
10^0
10^{-2}
10^{-4}
10^{-6}
10^0
10^1
10^2
10^3
Frequency (rad/s)
Phase (degrees)
200
100
0
-100
-200
10^0
10^1
10^2
10^3
Frequency (rad/s)

(c)



Original and Filtered Dash
Original
Filtered
30
20
10
0
-10
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
Original and Filtered Dot
Original
Filtered
10
0
-10
-20
-30
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5

(d)



Yes, the result perfectly aligns with theoretical expectations. The original `dash` signal is a baseband (low-frequency) signal that easily passes through the given lowpass filter. However, multiplying it by a high-frequency cosine wave modulates the signal, shifting its entire frequency spectrum to higher frequencies centered around $\pm f_1$. Since these new frequency components lie entirely outside the passband of the lowpass filter, they are attenuated and eliminated, resulting in an output that is practically zero.

(e)

Using Euler's formulas and trigonometric identities, we can determine the Fourier transforms. By applying the frequency-shifting property of the Fourier transform $(\cos(2\pi f_c t)m(t) \leftrightarrow \frac{1}{2}M(j(\omega - 2\pi f_c)) + \frac{1}{2}M(j(\omega + 2\pi f_c)))$, we derive the following:

Case 1: $m(t)\cos(2\pi f_1 t)\cos(2\pi f_1 t)$

Using the trigonometric identity $\cos^2(\theta) = \frac{1+\cos(2\theta)}{2}$:

$$\text{Signal} = \frac{1}{2}m(t) + \frac{1}{2}m(t)\cos(4\pi f_1 t) \tag{5}$$

Taking the Fourier transform:

$$\mathcal{F}\{\cdot\} = \frac{1}{2}M(j\omega) + \frac{1}{4}M(j(\omega - 4\pi f_1)) + \frac{1}{4}M(j(\omega + 4\pi f_1)) \tag{6}$$

Case 2: $m(t)\cos(2\pi f_1 t)\sin(2\pi f_1 t)$
Using the identity $\cos(\theta)\sin(\theta) = \frac{1}{2}\sin(2\theta)$:

$$\text{Signal} = \frac{1}{2}m(t)\sin(4\pi f_1 t) \tag{7}$$

Taking the Fourier transform:

$$\mathcal{F}\{\cdot\} = \frac{1}{4j}M(j(\omega - 4\pi f_1)) - \frac{1}{4j}M(j(\omega + 4\pi f_1)) \tag{8}$$

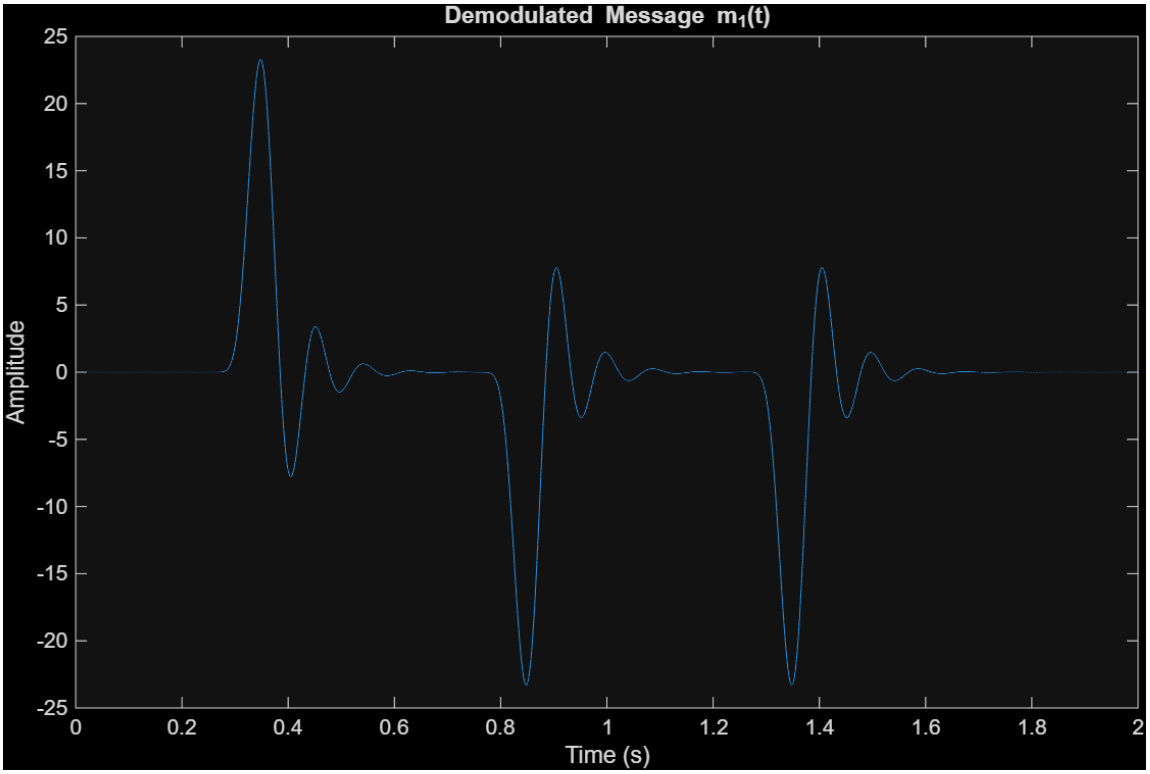
Case 3: $m(t)\cos(2\pi f_1 t)\cos(2\pi f_2 t)$
Using the identity $\cos(A)\cos(B) = \frac{1}{2}\cos(A+B) + \frac{1}{2}\cos(A-B)$:

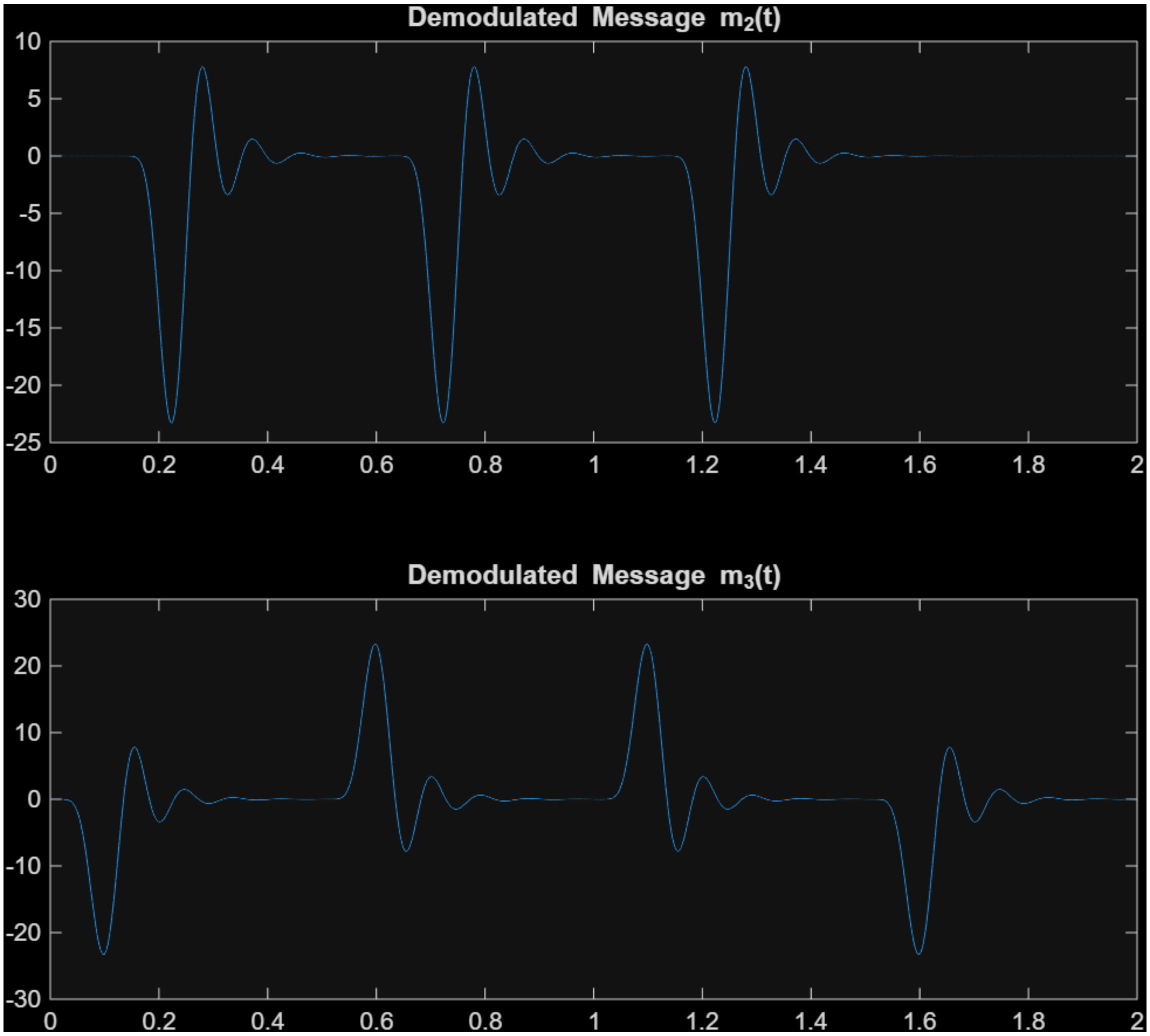
$$\text{Signal} = \frac{1}{2}m(t)\cos(2\pi(f_1 + f_2)t) + \frac{1}{2}m(t)\cos(2\pi(f_1 - f_2)t) \tag{9}$$

Taking the Fourier transform:

$$\begin{aligned}\mathcal{F}\{\cdot\} = {} & \frac{1}{4}M(j(\omega - 2\pi(f_1 + f_2))) + \frac{1}{4}M(j(\omega + 2\pi(f_1 + f_2))) \\ & + \frac{1}{4}M(j(\omega - 2\pi(f_1 - f_2))) + \frac{1}{4}M(j(\omega + 2\pi(f_1 - f_2)))\end{aligned} \tag{10}$$

(f)&(g)


Demodulated Message $m_1(t)$
Amplitude
Time (s)
25
20
15
10
5
0
-5
-10
-15
-20
-25
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2



Demodulation Strategy:
To extract a specific baseband message from the composite modulated signal $x(t)$, synchronous demodulation is required. We must multiply $x(t)$ by the exact carrier wave that was initially used to modulate the desired message.

For example, multiplying $x(t)$ by $\cos(2\pi f_1 t)$ isolates $m_1(t)$ at the baseband (producing a $\frac{1}{2}m_1(t)$ component), while simultaneously shifting the other messages (m_2 and m_3) to higher frequencies (e.g., $f_1 + f_2$). Passing this result through a lowpass filter removes all the unwanted high-frequency components. Finally, multiplying the filtered signal by 2 compensates for the amplitude attenuation caused by the trigonometric identities.

Results & Morse Code Decoding:
By applying this methodology in MATLAB and observing the plotted time-domain signals, we extract the following Morse codes:

- $m_1(t)$ represents the Morse code `-..`, which corresponds to the letter **D**.
- $m_2(t)$ represents the Morse code `...`, which corresponds to the letter **S**.
- $m_3(t)$ represents the Morse code `.--.`, which corresponds to the letter **P**.

Message to Agent 008:
The fully decoded message is the acronym **DSP**. Therefore, the aging Agent 007's final prediction is:

"The future of technology lies in ***DSP*** *(Digital Signal Processing)."*